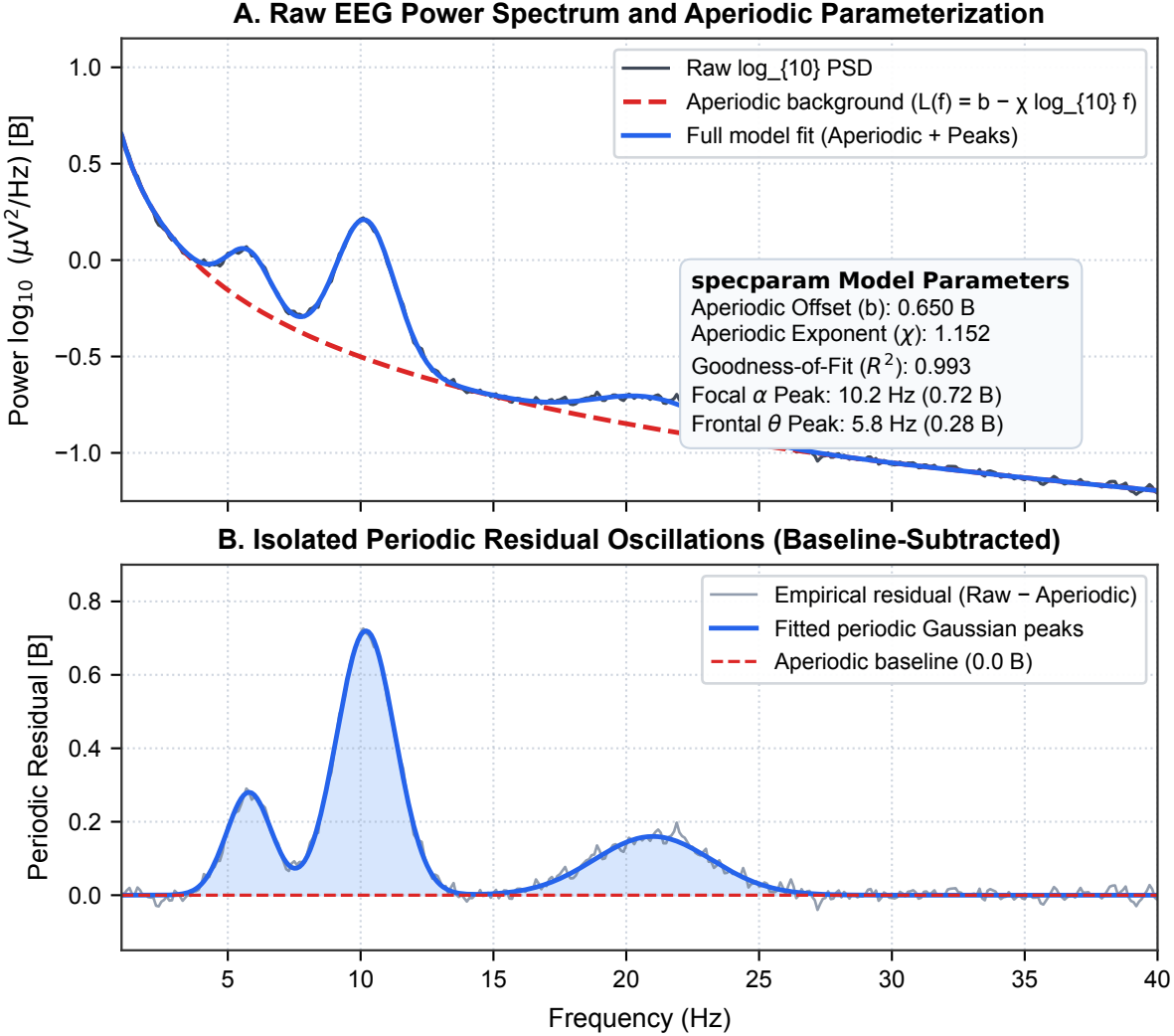

A. Raw EEG Power Spectrum and Aperiodic Parameterization
Raw log_{10} PSD
Aperiodic background (L(f) = b − χ log_{10} f)
Full model fit (Aperiodic + Peaks)
specparam Model Parameters
Aperiodic Offset (b): 0.650 B
Aperiodic Exponent (χ): 1.152
Goodness-of-Fit (R^2): 0.993
Focal α Peak: 10.2 Hz (0.72 B)
Frontal θ Peak: 5.8 Hz (0.28 B)
Power log$_{10}$ (μV^2/Hz) [B]
1.0
0.5
0.0
−0.5
−1.0
B. Isolated Periodic Residual Oscillations (Baseline-Subtracted)
Empirical residual (Raw − Aperiodic)
Fitted periodic Gaussian peaks
Aperiodic baseline (0.0 B)
Periodic Residual [B]
0.8
0.6
0.4
0.2
0.0
5
10
15
20
25
30
35
40
Frequency (Hz)